# Classifying Human Activity Using Sensor Data

Machine Learning Mini-Project | Sec-G | Team G-06 | Problem No.6

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## 1. Problem Statement

Classify the physical activity a person is performing (e.g. walking, running, cycling, ironing) from the signals of body-worn motion sensors, and compare several machine-learning models. The system must generalise to **people it has never seen**, so evaluation uses a subject-wise train/test split.

## 2. Dataset

**PAMAP2 Physical Activity Monitoring** (UCI Machine Learning Repository, ID 231). Nine subjects wore three inertial measurement units (IMUs) on the hand, chest and ankle, sampled at 100 Hz, plus a heart-rate monitor. Each row has 54 columns: timestamp, activity label, heart rate and 17 values per IMU (temperature, two accelerometers, gyroscope, magnetometer, orientation). We use the *Protocol* recordings, which contain **12 activities**: lying, sitting, standing, walking, running, cycling, Nordic walking, ascending stairs, descending stairs, vacuum cleaning, ironing, rope jumping. Subject 109 is excluded (its recording covers only a small subset of activities). Heart rate and orientation columns are excluded, leaving **39 sensor channels**.

## 3. Approach

- **Subject-wise split:** subjects 101-106 for training and 107-108 for testing, so no person appears in both sets; this avoids the optimistic results of random window splits.
- **Segmentation:** the signal is cut into contiguous single-activity runs. A run ends at an activity change, a removed (label 0) row, or a timestamp gap above 0.05 s, so windows never mix activities or span recording gaps.
- **Missing data:** gaps of at most 10 samples (0.1 s) are linearly interpolated inside a run only; windows with longer partial dropouts are skipped (3 of 19,201 windows were skipped).
- **Windows and features:** 2 s windows (200 samples) with 50 % overlap. For each channel we compute mean, standard deviation, min, max, median, IQR and RMS, giving 273 features per window (14,293 training and 4,908 test windows).
- **Models:** Random Forest, Extra Trees, Logistic Regression and Linear SVM, with class-balanced weights. Median imputation and scaling are inside each pipeline and fitted on training data only.
- **Model selection:** subject-grouped cross-validation on the training subjects only (macro-F1). The test subjects are never used to fit, tune or choose a model.

## 4. Implementation Overview

Python 3 (pandas, NumPy, scikit-learn, Matplotlib, joblib) in one script, `pamap2_har_complete.py`, run in Google Colab. It downloads and validates the dataset (retries, ZIP integrity check, extracts only the needed Protocol files), builds windows and features, trains and compares the four models, and writes `model_comparison.csv`, a classification report, confusion matrix, feature importances, the saved model and test predictions. The full run takes about 3 minutes. `demo_predict.py` loads the saved model and classifies real windows from a held-out subject for live demonstration. The code was validated with unit tests on synthetic data covering windowing, alignment, interpolation and download recovery before the final run on the real dataset.

## 5. Results

| Model | CV macro-F1 (train subj.) | Test accuracy | Test balanced acc. | Test macro-F1 |
|---|---|---|---|---|
| Extra Trees (selected) | 0.778 ± 0.085 | 0.934 | 0.915 | 0.927 |
| Random Forest | 0.733 ± 0.038 | 0.872 | 0.861 | 0.869 |
| Logistic Regression | 0.667 ± 0.034 | 0.715 | 0.740 | 0.677 |
| Linear SVM | 0.578 ± 0.020 | 0.685 | 0.694 | 0.663 |

Per-class F1 of the selected model (test subjects): Lying 0.99, Cycling 0.99, Nordic Walking 0.98, Walking 0.96, Sitting 0.93, Running 0.93, Ascending Stairs 0.93, Ironing 0.90, Standing 0.89, Descending Stairs 0.88, Rope Jumping 0.88,
Vacuum Cleaning 0.86.

## 6. Conclusions

Extra Trees was the best model, reaching **93.4 % accuracy and 0.927 macro-F1** on unseen subjects; it also ranked first in cross-validation, so the choice was made without using the test set. Tree ensembles clearly outperformed the linear models (about 0.93 vs 0.66-0.68 macro-F1), showing that the relationship between simple window statistics and activity is non-linear. Distinctive, high-motion activities (lying, cycling, Nordic walking, walking) are recognised almost perfectly, while low-intensity household activities (vacuum cleaning, ironing) and standing are the hardest because they produce similar, mild body motion: the most frequent errors are ironing predicted as vacuum cleaning (58 windows) and the reverse (34), and several activities (descending stairs, vacuum cleaning, ironing) are sometimes predicted as standing. The

cross-validation score (0.78 ± 0.085) is a more conservative estimate than the test score, because each fold trains on fewer subjects and the test set contains only two subjects.

**Limitations / future work:** only two test subjects; no hyperparameter tuning; temperature channels may carry session-specific information. Future work could use leave-one-subject-out evaluation, deep models (CNN/LSTM) on raw windows, and sensor-subset analysis for fewer wearables.